

2. Two-dimensional Array:

Arrays that require two indexes to identify a particular element.

The following figure shows two-dimensional (2D) array named `a` contains 3 rows and 4 columns:

	Column 0	Column 1	Column 2	Column 3
Row 0	<code>a[0][0]</code>	<code>a[0][1]</code>	<code>a[0][2]</code>	<code>a[0][3]</code>
Row 1	<code>a[1][0]</code>	<code>a[1][1]</code>	<code>a[1][2]</code>	<code>a[1][3]</code>
Row 2	<code>a[2][0]</code>	<code>a[2][1]</code>	<code>a[2][2]</code>	<code>a[2][3]</code>

Column index
Row index
Array name

Note:

- The indexes of the array start with *zero* always.
- The lengths of the rows in 2D array are *not* required to be the same.
- Array could extend naturally to 3D, 4D, and even higher-dimensional, but they are not used very often in practice.

How to declare, initialize and access 2D array?

The syntax for creating 2D Array as following:

```
data_type array_name[][];
```

OR

```
data_type[][] array_name;
```

```
int[][] array; // Declare 2D array of type int named "array"  
int array[][]; // Same as above
```

Specifying a 2D array dimensions:

```
double[][] c = new double[4][5]; // Declare an double 2D array named  
"c" with 4 rows and 5 columns  
double c[][] = new double[4][5]; // Same as above
```

Like one-dimensional arrays, 2D arrays could be declare and initialize:

```
int[][] A = {
    {11, -25, 4, 77},
    {-21, 55, 43, 11}
}; // This creates and initializes a 2D array of int
with 2 rows and 4 columns

int[][] A = new int[2][4]; // Same as above
// Initialize first row of values
A[0][0] = 11;
A[0][1] = -25;
A[0][2] = 4;
A[0][3] = 77;
// Initialize second row of values
A[1][0] = -21;
A[1][1] = 55;
A[1][2] = 43;
A[1][3] = 11;
```

A two-dimensional array with a variable number of columns is called a *ragged array*. For example:

```
char[][] animals = {
    {'M', 'O', 'N', 'K', 'E', 'Y'}, // 6 columns
    {'C', 'A', 'T'}, // 3 columns
    {'B', 'I', 'R', 'D'} // 4 columns
}; // this creates a 2D array with a variable
number of columns
```

0	'M'	'O'	'N'	'K'	'E'	'Y'
1	'C'	'A'	'T'			
2	'B'	'I'	'R'	'D'		
	0	1	2	3	4	5

We could use the *length attribute* to get array information:

```
int array[][] = new int[12][8];    // a 12x8 array of int
System.out.println(array.length);  // 12 number of rows
System.out.println(array[0].length); // 8 number of columns
System.out.println(array[11].length); // 8 number of columns
```

To access an element in a 2D array requires both a row and a column index. And a *pair* of nested loops is usually used to process each element.

```
public class Array2DExample1 {
    public static void main(String args[]) {
        // Declaring and initializing 2D array
        int arr[][] = { {1,2,3},{4,5,6},{7,8,9} };
        // Printing 2D array
        for (int i=0; i< 3; i++)
        {
            for (int j=0; j< 3; j++)
            {
                System.out.print(arr[i][j] + " ");
            }
        }
    }
}
```



```
public class Array2DExample2 {
    public static void main(String args[]) {
        // Declaring and initializing ragged array
        char[][] animals= {
            {'M', 'O', 'N', 'K', 'E', 'Y'},
            {'C', 'A', 'T'},
            {'B', 'I', 'R', 'D'}
        };
        for (int row = 0; row < animals.length; row++)
        {
            for (int col = 0; col < animals[row].length; col++)
            {
                System.out.print(animals[row][col]); // Display one character
            }
            System.out.println(); // New line after one row display
        }
    }
}
```



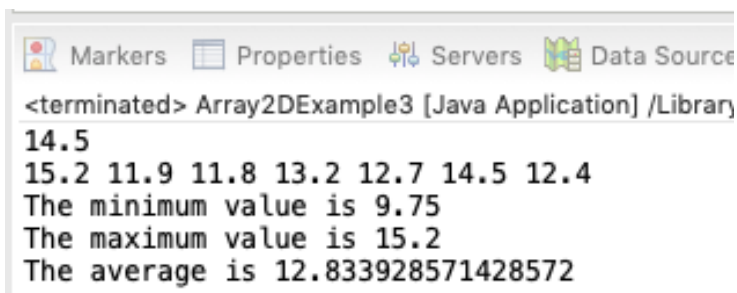
Example 3: Consider the following 2D array named temperatures:

	0	1	2	3	4	5	6
0	12.5	12.2	14.0	13.0	13.2	13.0	12.2
1	15.2	11.9	11.8	13.2	12.7	14.5	12.4
2	9.75	11.8	11.9	13.5	12.7	14.7	13.0
3	14.0	13.2	11.8	14.0	14.3	11.0	11.9

Perform the following:

1. Print out the value of highlighted indexes.
2. Print out the value of highlighted row.
3. Find out the maximum, minimum and average values.

```
public class Array2DExample3 {  
    public static void main(String args[]) {  
        double[][] temperatures = {{12.5,12.2,14.0,13.0,13.2,13.0,12.2},  
                                    {15.2,11.9,11.8,13.2,12.7,14.5,12.4},  
                                    {9.75,11.8,11.9,13.5,12.7,14.7,13.0},  
                                    {14.0,13.2,11.8,14.0,14.3,11.0,11.9}};  
  
        double min,max,sum;  
        min= temperatures[0][0]; // Assume the first item is the minimum  
        max= temperatures[0][0]; // Assume the first item is the maximum  
        sum=0;  
        System.out.println(temperatures[1][5]);  
  
        for (int col=0; col< temperatures[1].length; col++)  
            System.out.print(temperatures[1][col] + " ");  
        System.out.println();  
  
        for (int row=0; row< temperatures.length; row++)  
        {  
            for (int col=0; col< temperatures[row].length; col++)  
            {  
                if (min > temperatures[row][col])  
                    min= temperatures[row][col];  
                if (max < temperatures[row][col])  
                    max= temperatures[row][col];  
                sum=sum+ temperatures[row][col];  
            }  
        }  
        System.out.println("The minimum value is " + min);  
        System.out.println("The maximum value is " + max);  
        System.out.println("The average is " + sum/28);  
    }  
}
```



```
<terminated> Array2DExample3 [Java Application] /Library  
14.5  
15.2 11.9 11.8 13.2 12.7 14.5 12.4  
The minimum value is 9.75  
The maximum value is 15.2  
The average is 12.833928571428572
```

Example 4: Write down a Java program to read a 2-dimensional array x[4][4] then perform each of the following:

1. Find the maximum and minimum in the third row
2. Replace the two diagonals with 1

```
import java.util.Scanner;
public class Array2DExample4 {
    public static void main(String args[]) {
        int [][] x = new int[4][4];
        int min,max;
        Scanner s = new Scanner(System.in);
        System.out.print("Enter the array elements: ");
        for (int i=0; i< 4; i++)
            for (int j=0; j< 4; j++)
                x[i][j] = s.nextInt();
        s.close();

        for (int i=0; i< 4; i++)
        {
            for (int j=0; j< 4; j++)
                System.out.print(x[i][j] + " ");
            System.out.println();
        }
        min= x[3][0]; // Assume the first item is the minimum
        max= x[3][0]; // Assume the first item is the maximum

        for (int j=0; j< 4; j++)
        {
            if (min > x[3][j])
                min= x[3][j];
            if (max < x[3][j])
                max= x[3][j];
        }
        System.out.println("The minimum value is " + min);
        System.out.println("The maximum value is " + max);

        for (int i=0; i< 4; i++)
        for (int j=0; j< 4; j++)
        {
            if (i == j) // Condition for principal diagonal
                x[i][j] = 1;
            if ((i+j) == 3) // Condition for secondary diagonal
                x[i][j] = 1;
        }
        for (int i=0; i< 4; i++)
        {
            for (int j=0; j< 4; j++)
                System.out.print(x[i][j] + " ");
            System.out.println();
        }
    }
}
```

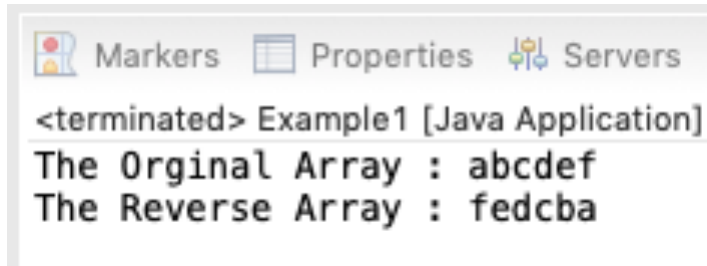
```
Markers Properties Servers Data Source Explorer Snippets Console
<terminated> Array2DExample4 [Java Application] /Library/Java/JavaVirtualMachines/jdk1.8.0_112.jdk/
Enter the array elements: 22 12 23 24 44 13 32 34 55 18 17 90 12 14 78 55
22 12 23 24
44 13 32 34
55 18 17 90
12 14 78 55
The minimum value is 12
The maximum value is 78
1 12 23 1
44 1 1 34
55 1 1 90
1 14 78 1
```

Homework:

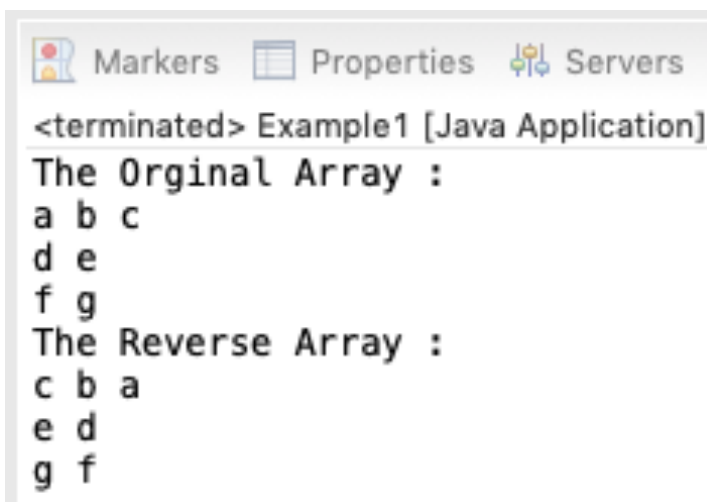
- Q1>** Write a Java program to count the number of occurrences of an element in an 2D array.
- Q2>** Write a Java program to print negative numbers in an 2D array.
- Q3>** Write a Java program to read of an 2D array then to print the two diagonals elements only.

Example1: Write a Java program to print an array elements in a reverse order? for example: Array1 = "abcdef" the output will be "fedcba".

```
public class Example1 {  
    public static void main(String[] args) {  
        char[] array= {'a', 'b', 'c', 'd', 'e', 'f'};  
        System.out.print("The Original Array : ");  
        for (int i = 0; i < array.length; i++)  
            System.out.print(array[i]);  
        System.out.print("\nThe Reverse Array : ");  
        for (int i = array.length-1; i >= 0; i--)  
            System.out.print(array[i]);  
    }  
}
```

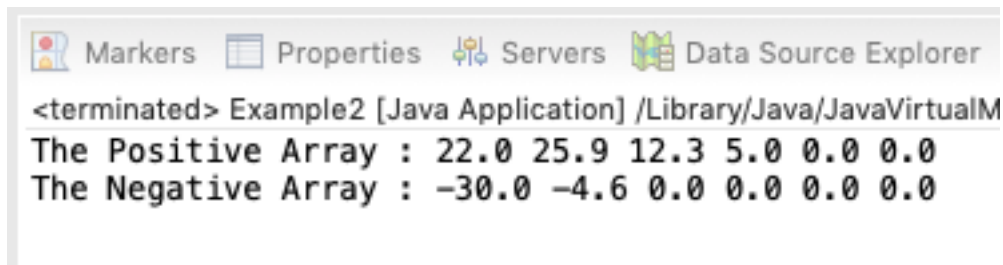


```
public class Example1 {  
    public static void main(String[] args) {  
        char[][] array1 = {{'a', 'b', 'c'}, {'d', 'e'}, {'f', 'g'}};  
        System.out.print("The Original Array : \n");  
        for (int i = 0; i < array1.length; i++)  
        {  
            for (int j=0; j < array1[i].length ; j++)  
            {  
                System.out.print(array1[i][j] + " ");  
            }  
            System.out.println();  
        }  
        System.out.print("The Reverse Array : \n");  
        for (int i = 0; i < array1.length; i++)  
        {  
            for (int j=array1[i].length-1; j >=0 ; j--)  
            {  
                System.out.print(array1[i][j] + " ");  
            }  
            System.out.println();  
        }  
    }  
}
```



Example2: Write a Java program to read an array of type double then store their positive value in an array and the negative numbers in another array?

```
public class Example2 {  
    public static void main(String[] args) {  
        double[] arr = {22, -30, 25.9, 12.3, -4.6, 5.0};  
        int n = arr.length;  
        double[] arr1 = new double[n];  
        double[] arr2 = new double[n];  
        int j=0,c=0;  
        for (int i = 0;i < n; i++)  
        {  
            if (arr[i] > 0)  
            {  
                arr1[j]=arr[i];  
                j++;  
            }  
            else {  
                arr2[c]=arr[i];  
                c++;  
            }  
        }  
        System.out.print("The Positive Array : ");  
        for (int i = 0;i < arr1.length; i++)  
        System.out.print(arr1[i] +" ");  
        System.out.print("\nThe Negative Array : ");  
        for (int i = 0;i < arr2.length; i++)  
        System.out.print(arr2[i] +" ");  
    }  
}
```



```
<terminated> Example2 [Java Application] /Library/Java/JavaVirtualM  
The Positive Array : 22.0 25.9 12.3 5.0 0.0 0.0  
The Negative Array : -30.0 -4.6 0.0 0.0 0.0 0.0
```

Example3: Write a Java program to find the summation of two integer arrays?

```
public class Example3 {  
    public static void main(String args[]) {  
        int[] array1 = { 20, 14, 6 };  
        int[] array2 = { 18, 30, 25 };  
        int[] result = new int[array1.length];  
  
        for (int i = 0;i < array1.length; i++)  
            result[i]= array1[i]+array2[i];  
  
        System.out.print("The Summation Array : ");  
        for (int i = 0;i < result.length; i++)  
            System.out.print(result[i] +" ");  
    }  
}
```



```
<terminated> Example3 [Java Application] /Library/J  
The Summation Array : 38 44 31
```


$$\begin{bmatrix} 3 & 8 \\ 4 & 6 \end{bmatrix} + \begin{bmatrix} 4 & 0 \\ 1 & -9 \end{bmatrix} = \begin{bmatrix} 7 & 8 \\ 5 & -3 \end{bmatrix}$$

3+4=7

```
public class Example3 {
    public static void main(String args[]) {
        int[][] array1 = {{3, 8}, {4, 6}};
        int[][] array2 = {{4, 0}, {1, -9}};
        int[][] result = new int[array1.length][array1.length];

        for (int i = 0; i < array1.length; i++)
            for (int j = 0; j < array1[i].length; j++)
                result[i][j] = array1[i][j] + array2[i][j];

        System.out.print("The Summation Array : \n");
        for (int i = 0; i < result.length; i++)
        {
            for (int j = 0; j < result[i].length; j++)
            {
                System.out.print(result[i][j] + " ");
            }
            System.out.println();
        }
    }
}
```

```
<terminated> Example3 [Java Application] /Library/J
The Summation Array :
7 8
5 -3
```

Note: Both examples the two arrays of **same type** and **same length**

Homework:

Q1> Write down a Java program to find the multiplication of two 2D arrays?

$$\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \begin{bmatrix} 2 & 0 \\ 1 & 2 \end{bmatrix} = \begin{bmatrix} 1 \times 2 + 2 \times 1 & 1 \times 0 + 2 \times 2 \\ 3 \times 2 + 4 \times 1 & 3 \times 0 + 4 \times 2 \end{bmatrix} = \begin{bmatrix} 4 & 4 \\ 10 & 8 \end{bmatrix}$$

Q2> Write down a Java program to print the index of even numbers in an array?

Q3> Write down a Java program that will double the value of every item in an array?

Q4> Write down a Java program to reverse the order of elements in an array?
for example: arr1 = {12, 30, 15, -4, 7} becomes arr1 = {7, -4, 15, 30, 12}.